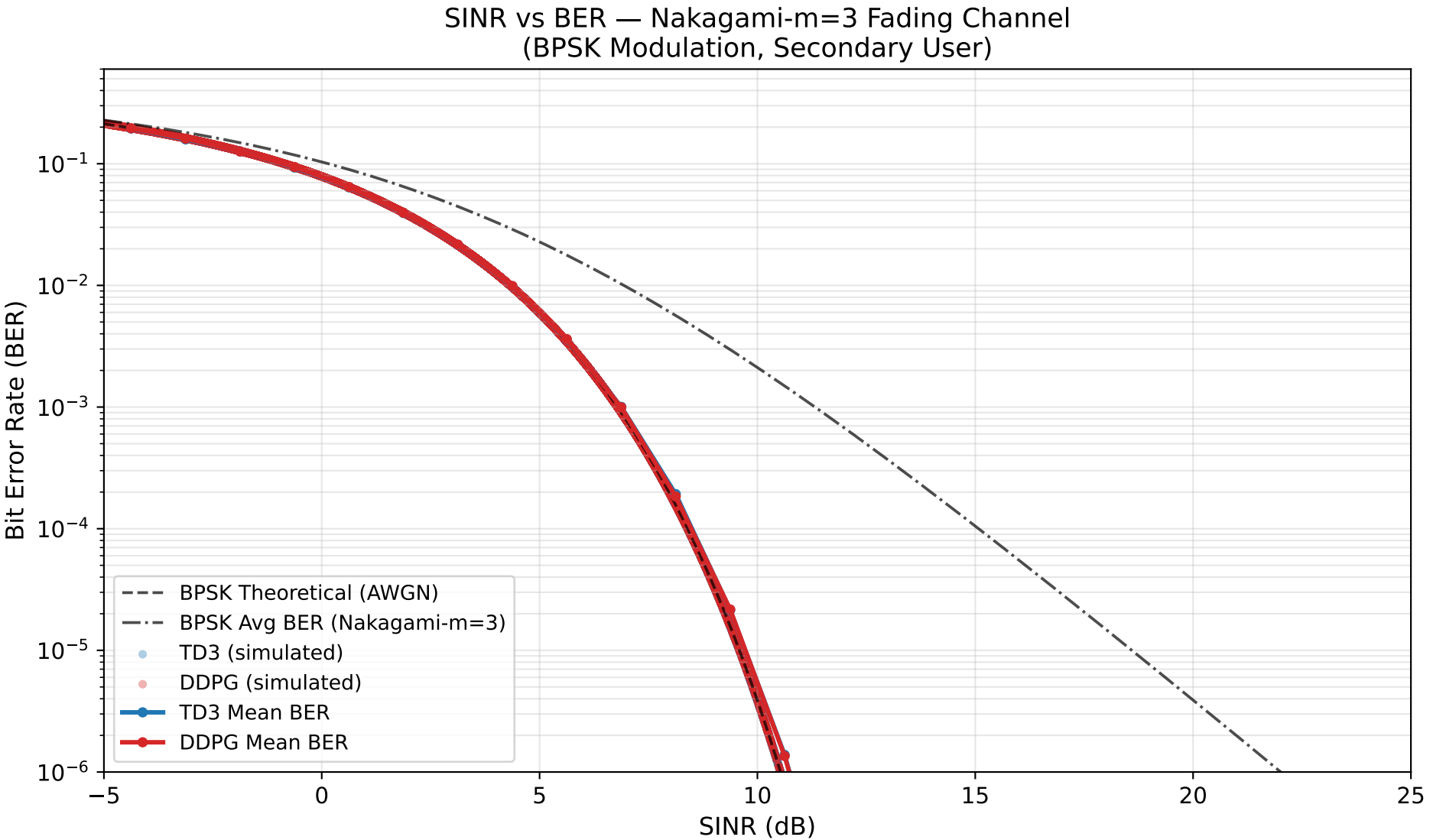
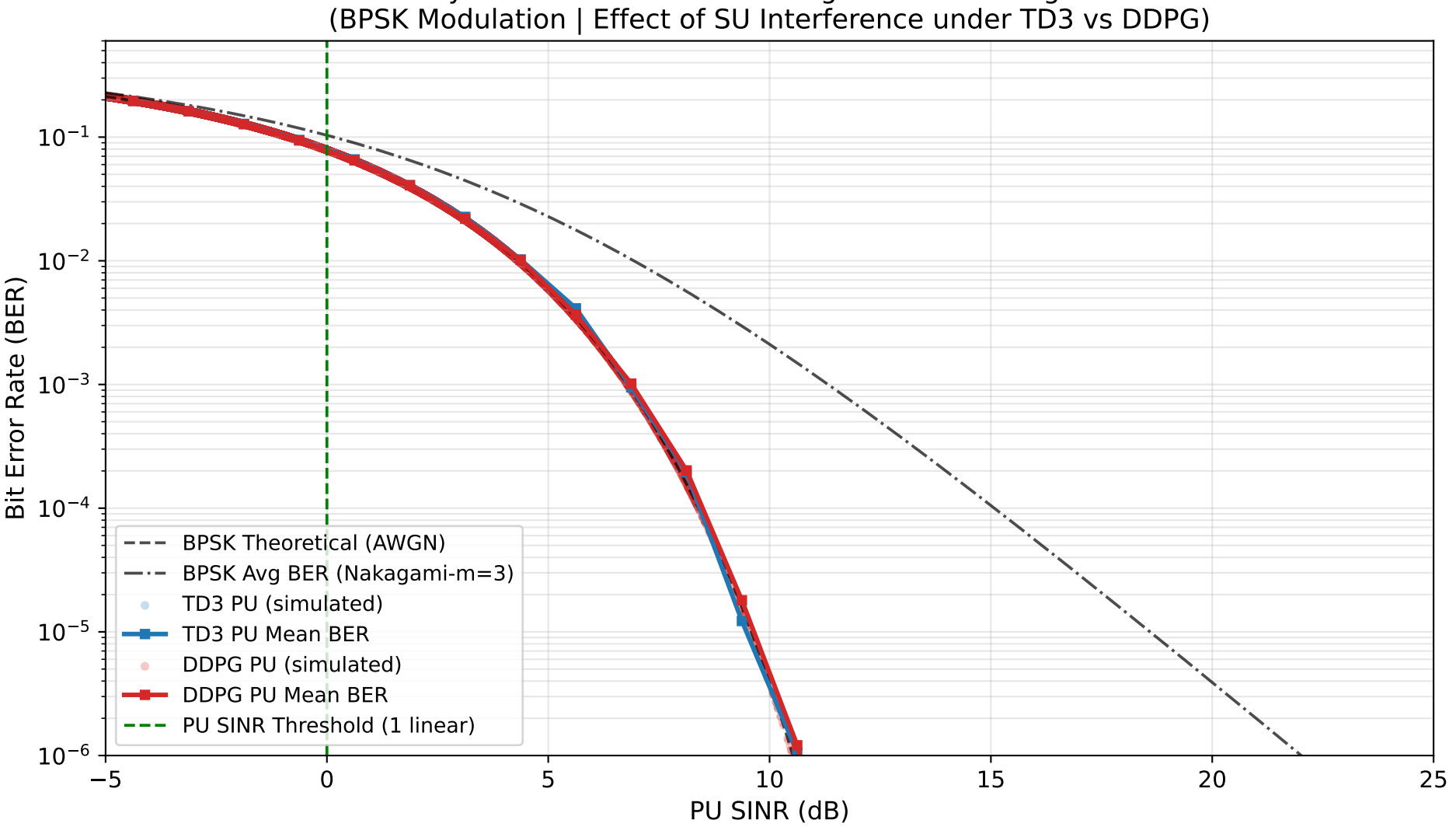
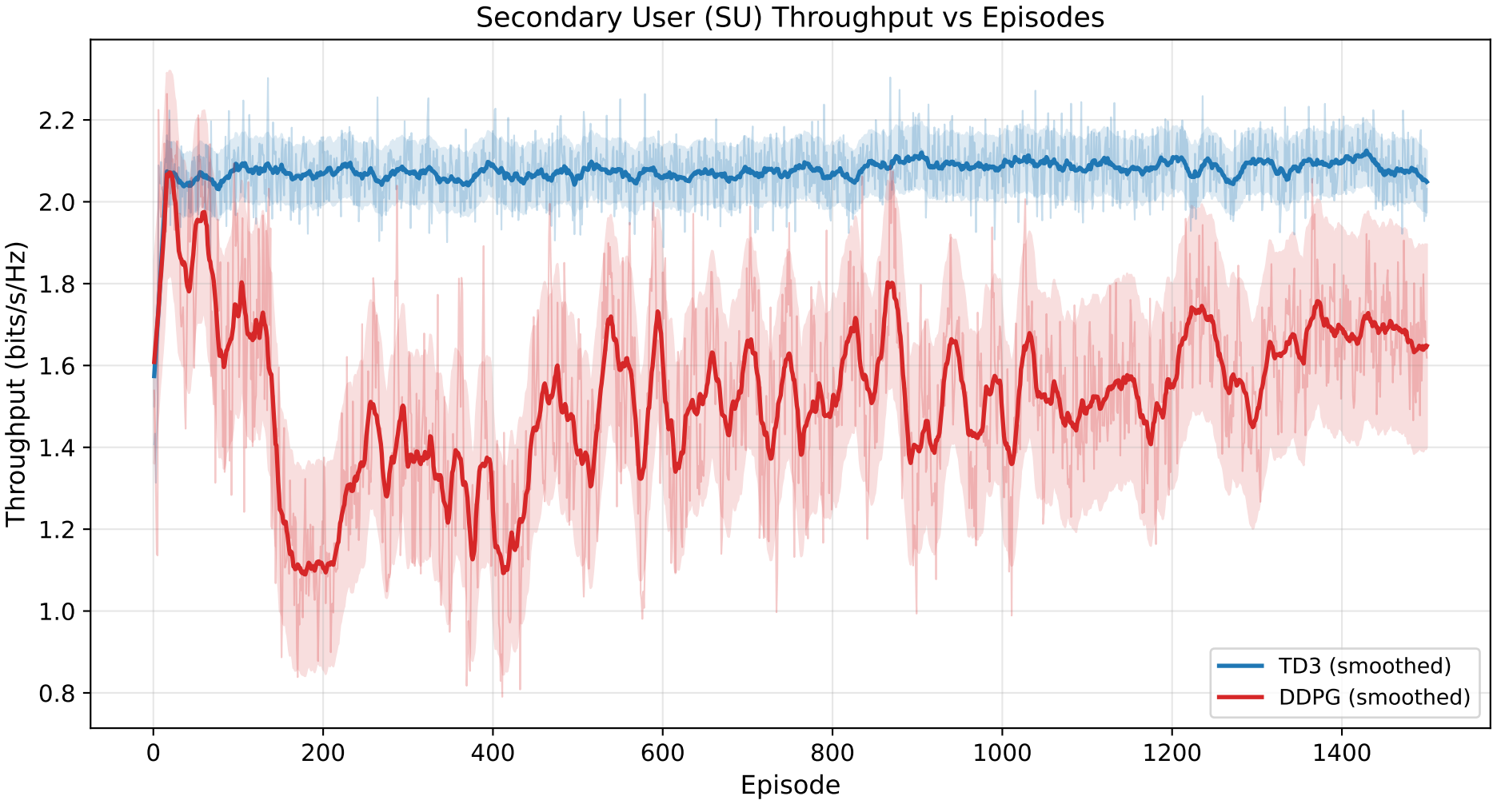
CRN Power Control: TD3 vs DDPG
Nakagami-m Fading Channel Performance Report

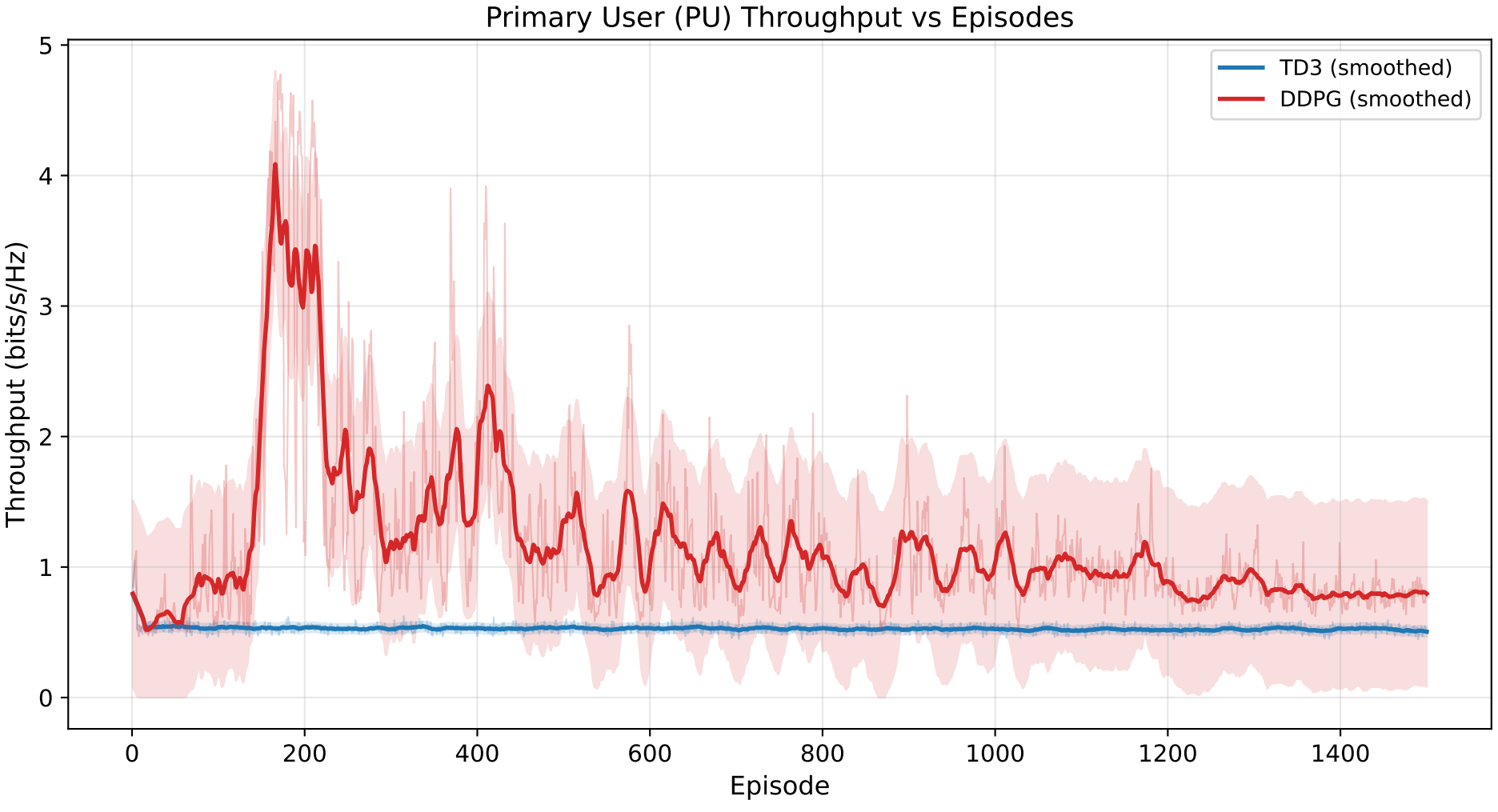
Nakagami-m = 3.0 | Episodes = 1500 | Steps/Episode = 200 | SINR Threshold = 1.0 dB

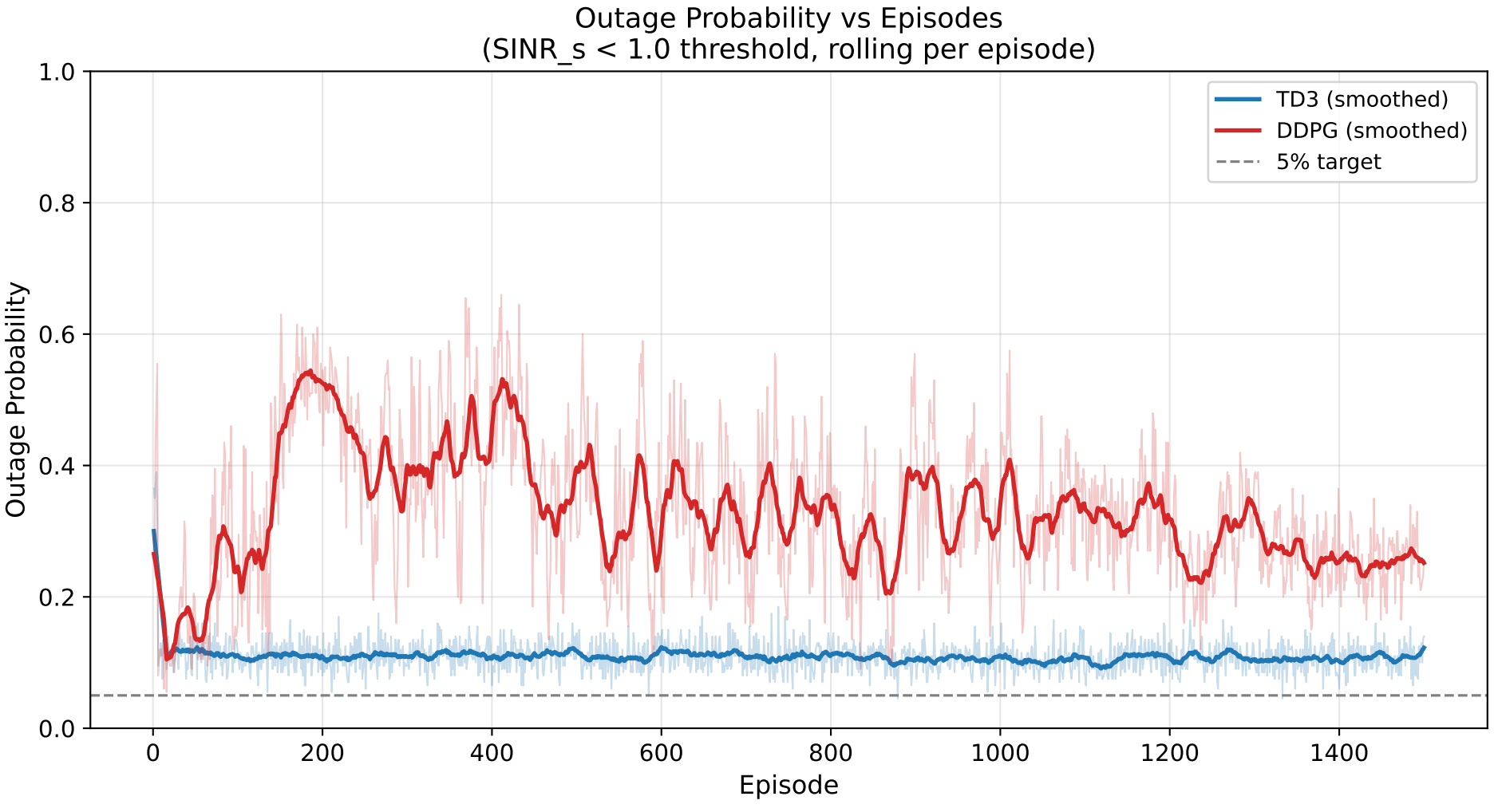
Metric	TD3	DDPG	Winner
Avg Reward (last 100 ep)	243.0847	205.6259	**TD3**
SU Throughput (bits/s/Hz)	2.0860	1.6818	**TD3**
PU Throughput (bits/s/Hz)	0.5222	0.7856	**DDPG**
Outage Probability	0.1084	0.2520	**TD3**
Average BER	0.027201	0.055410	**TD3**
Training Time (s)	6448.6	5960.9	—

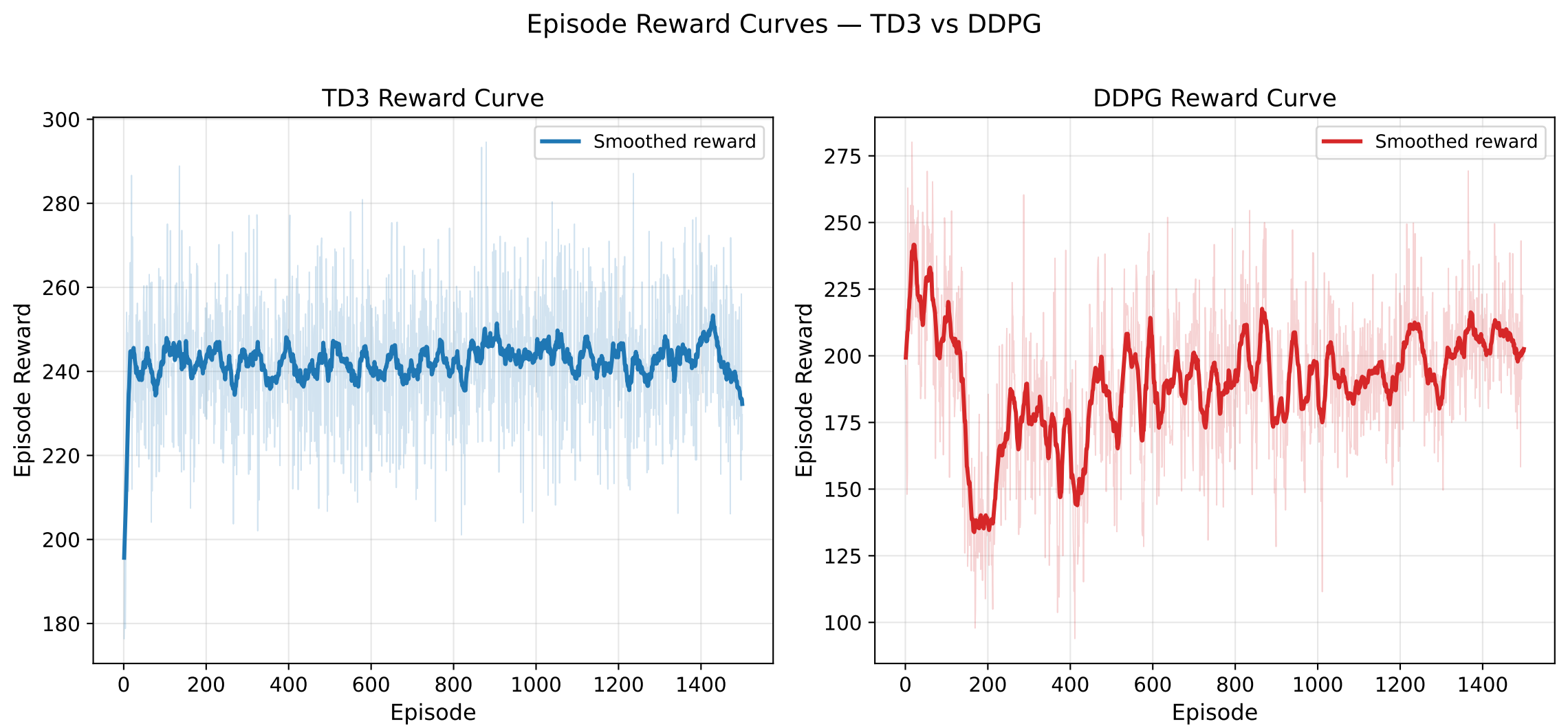

SINR vs BER — Nakagami-m=3 Fading Channel
(BPSK Modulation, Secondary User)
Bit Error Rate (BER)
SINR (dB)
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
10^{-6}
−5
0
5
10
15
20
25
BPSK Theoretical (AWGN)
BPSK Avg BER (Nakagami-m=3)
TD3 (simulated)
DDPG (simulated)
TD3 Mean BER
DDPG Mean BER

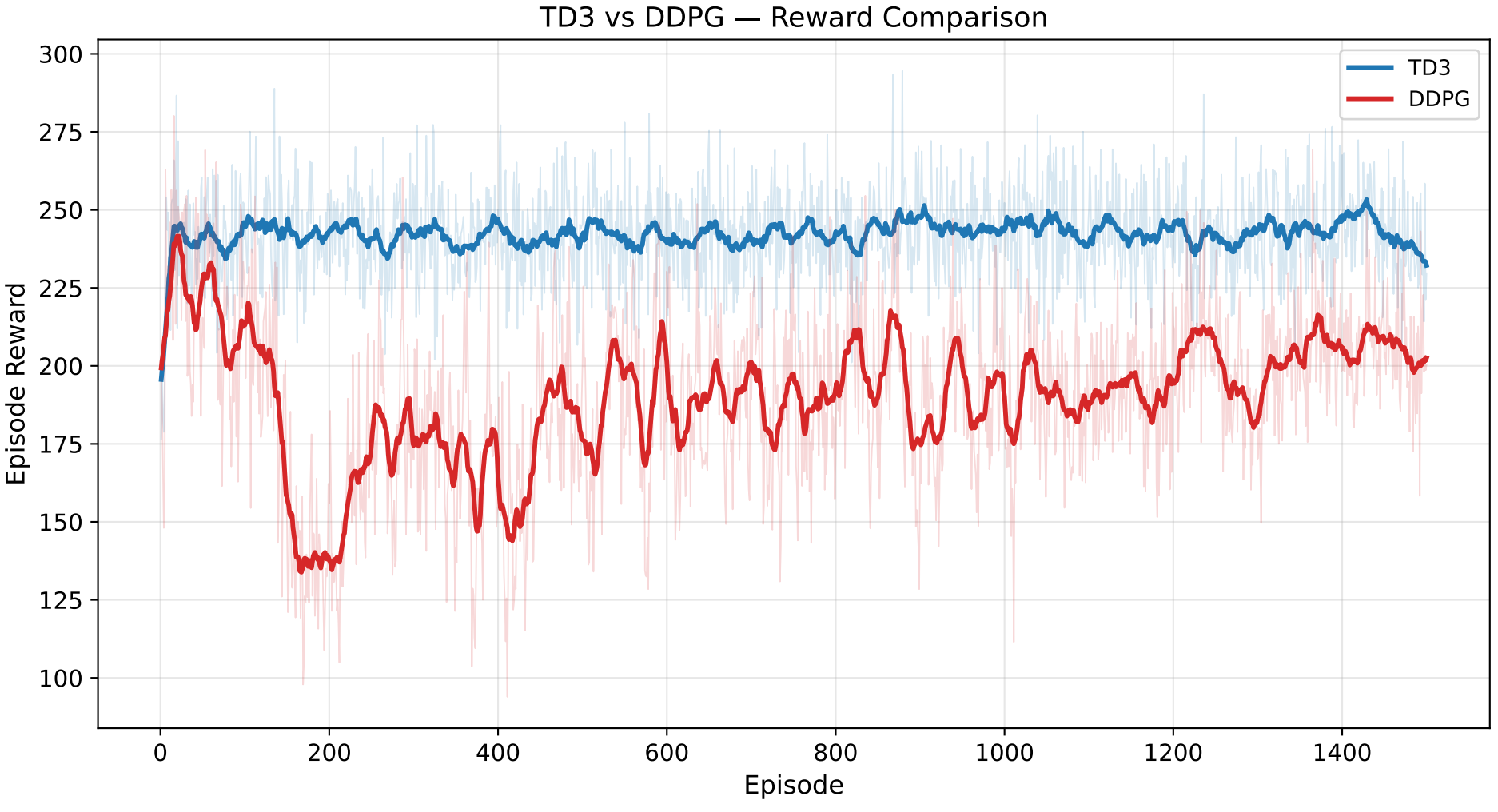

Primary User SINR vs BER — Nakagami-m=3 Fading Channel
(BPSK Modulation | Effect of SU Interference under TD3 vs DDPG)
Bit Error Rate (BER)
PU SINR (dB)
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
10^{-6}
−5
0
5
10
15
20
25
BPSK Theoretical (AWGN)
BPSK Avg BER (Nakagami-m=3)
TD3 PU (simulated)
TD3 PU Mean BER
DDPG PU (simulated)
DDPG PU Mean BER
PU SINR Threshold (1 linear)


Secondary User (SU) Throughput vs Episodes
Throughput (bits/s/Hz)
Episode
0.8
1.0
1.2
1.4
1.6
1.8
2.0
2.2
0
200
400
600
800
1000
1200
1400
TD3 (smoothed)
DDPG (smoothed)


Primary User (PU) Throughput vs Episodes
TD3 (smoothed)
DDPG (smoothed)
Throughput (bits/s/Hz)
Episode
0
1
2
3
4
5
0
200
400
600
800
1000
1200
1400


Outage Probability vs Episodes
(SINR_s < 1.0 threshold, rolling per episode)
TD3 (smoothed)
DDPG (smoothed)
5% target
Outage Probability
Episode
0.0
0.2
0.4
0.6
0.8
1.0
0
200
400
600
800
1000
1200
1400


Episode Reward Curves — TD3 vs DDPG
TD3 Reward Curve
Smoothed reward
Episode Reward
Episode
300
280
260
240
220
200
180
0
200
400
600
800
1000
1200
1400
DDPG Reward Curve
Smoothed reward
Episode Reward
Episode
275
250
225
200
175
150
125
100
0
200
400
600
800
1000
1200
1400


TD3 vs DDPG — Reward Comparison
TD3
DDPG
Episode Reward
Episode
300
275
250
225
200
175
150
125
100
0
200
400
600
800
1000
1200
1400